

intensity of LED connected to these pins.

The below sketch uses the PWM Pin 3 to achieve light intensity control for the LED connected to Pin 3.

In the above code, the LED connected to Pin 3 is given a varying voltage that looks like the PWM wave in the given figure.



PWM for LED light intensity via voltage control

The average voltage output of analog pin can be calculated by the following formula:

Thus, putting the above value of 64 gives voltage of 1.25 volts. i.e. 25% of 5 V, thereby giving 25% light intensity.

Send and receive data with computer via Serial Communication

Though LED can help during debugging process, they can only help monitor one condition for a specific thing. To monitor multiple parameters on the Arduino™ or to give instructions, we need to receive or send data using the serial port of the computer.

Data transfer in either direction is possible via the USB cable that connects the Arduino™ and the computer.

<code>void setup () { Serial.begin (9600); }</code>	Start the serial communication interface on the Arduino™ setting the data transfer rate to 9600 bits per second or 9.6 kbps.
<code>void loop ()</code>	
<code>{</code>	
<code> if (Serial.available ())</code>	Check if Arduino™ has received serial data. Else you'll be reading false values.
<code> {</code>	
<code> char data = Serial.read ();</code>	If data is available read the data.
<code> Serial.print ("You entered: ");</code>	Send the text "You entered: " on the serial port. The text is visible on a serial monitor .
<code> Serial.println (data);</code>	The Serial.println () puts a line feed (next line), after display the data in.
<code> }</code>	
<code>}</code>	